How Can Energy Transformations Solve Problems?

Chapter 6

Big Question

How do energy transformations help people?

Imagine living in a time before toasters. You want to toast a piece of bread. You hold the bread over a fire, but it burns too quickly. Now you have burnt bread! This is a problem. Suppose you need to meet a friend in the morning, but alarm clocks have not been invented yet. You are late, and your friend leaves without you. This is a problem!

Engineers are people who use science and technology to make things that people want or need. To an engineer, anything that people want or need presents a problem to be solved. All kinds of problems can be solved through the design of new devices. Toasters and alarm clocks are devices that help people. Many such devices are designed to make use of energy transfers and transformations.



Problem: You want to heat bread to just the right temperature.

Solution: The toaster changes electrical energy to heat, motion, and sound energy. The toasty bread pops up, *DING*, at just the right time.

Solving Problems Requires Understanding Energy

In finding solutions to problems, engineers consider the properties of energy. They know that energy can transfer, or move, from place to place. As a bowling ball collides with bowling pins, some of the ball's energy of motion is transferred to the pins. Your score goes up. Problem solved.

Engineers also know, as do you, that energy can transform, or **convert**, from one form to another. When a bowling ball collides with pins, some of its energy of motion converts to sound energy as the surrounding air vibrates. Such **conversions** occur when energy changes from one form to another.

People use what they know about energy changes to design and build devices that make use of these **energy transformations**. These devices can make life easier, more productive, or even just more fun!



Vocabulary

convert, v. to change form

conversion, n. the process of being converted or changing form

energy transformation, n. the change of one form of energy to another form of energy

Problem: You need to know what time it is.

Solution: Mechanical clocks use springs to store energy. They transform stored energy to motion energy. The moving clock hands show people the time of day.

Transforming Motion Energy Can Help Solve Problems

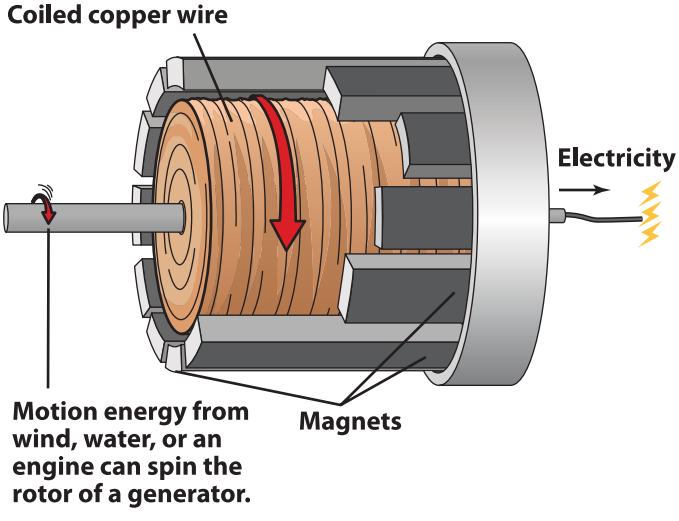
People need electrical energy to power all sorts of devices, but where does the electrical energy come from? Most of the electricity we use in our homes comes from the transformation of motion energy to electrical energy.

In 1831 a scientist named Michael Faraday discovered that moving magnets past copper wires in certain ways produces a flow of electrical energy through the wires. He discovered a way of transforming motion energy to electrical energy.

Many engineers have used Faraday's discovery to build generators, rotating devices that convert motion to electricity. Hydroelectric dams and the turbines on wind farms use the movement of water and wind to rotate generators. The spinning motion in the generators is converted to electrical energy.

Word to Know

Rotate means to spin. For example, blades of a ceiling fan rotate around and around to make air move in a room.



Problem: People need electricity to power many of the devices they use.

Solution: Generators convert motion energy from spinning parts to electrical energy.

Transforming Stored Energy Can Help Solve Problems

Batteries provide electricity, too. Batteries are made of chemicals that have stored energy. The stored energy in a battery transforms to electrical energy. In turn, that electrical energy can be transformed in useful ways by many other devices.

Some batteries "die" after all their stored energy has been transformed to electrical energy. Other batteries can be recharged. Rechargeable batteries can convert electricity from a wall socket to stored chemical energy for later use.

Rechargeable batteries are especially useful for small, portable devices. People want to carry some electronic devices with them wherever they go. These electronic devices use energy transformations in several ways.

A cell phone, for example, converts stored energy in its battery to electrical energy. The electrical energy from the battery transforms to

- sound energy when the phone rings and voices come over the speaker,
- light energy when the screen displays pictures or videos, and
- motion energy when the phone is set to vibrate for notifications.



Problem: People want to be connected to others wherever they go.

Solution: Smartphones with rechargeable batteries store electricity and transform it to sound, light, and motion.

Transforming Light Energy Can Help Solve Problems

Solar cells are made of material that transforms sunlight to electrical energy. The transformation of energy from sunlight to electrical energy also happens on a very large scale. Solar panels convert sunlight to electrical energy during the day. Some of that electrical energy can be stored in batteries to power useful devices at night when the sun is not out.



The solar pad that can charge this phone's battery is small and portable.



A bigger solar panel can convert more sunlight to electrical energy. It can provide electricity for a home that does not have access to electricity generated by a power plant.



Enough solar panels in a sunny region can provide all the electricity needed by an entire community. It doesn't have to be hot to be sunny!



You don't even have to be on Earth for it to be sunny, either! Solar panels solve the problem of how to get the electricity needed to operate the International Space Station.

Engineering Design Solves Problems

All the devices that you have just read about transform energy. They have something else in common, too. They were developed using a process called **engineering design**. Engineering design is used by scientists and engineers to develop solutions to problems.

Vocabulary

engineering design, n. a process used to develop a solution to a problem

The process starts with identifying a problem to be solved. Teams of scientists and engineers determine a want or need that the design is going to meet. Then, they judge possible solutions by how practical they are and how well they solve the problem.

For example, electric cars must have rechargeable batteries that store a lot more electrical energy than a phone battery. An electric car battery must store enough energy but not be too heavy or too expensive. Designers have to consider things like cost and weight to solve problems without creating new ones.



Engineers who design electric cars must also design charging stations to transform electricity to stored energy in the cars' batteries.